# Model-Based Three-Dimensional Digital Image Vibration Measurement Method Using Stereo High-Speed Cameras

Wenxiang Qin (1), Kohei Shimasaki (1), Feiyue Wang (1), Shaopeng Hu (2), Idaku Ishii (1)
1 : Graduate School of Advanced Science and Engineering, Hiroshima University, Japan
2 : Digital Monozukuri (Manufacturing) Education and Research Center, Hiroshima University, Japan

**Abstract:** By introducing three-dimensional model, a three-dimensional vibration measurement method based on digital image correlation using stereo high-speed cameras is proposed to overcome the problems of conventional methods. An experiment using a speaker verified its effectiveness.
© 2024 OSJ

**Keywords:** digital image correlation, vibration measurement, three-dimensional model, stereo cameras, high speed.

## 1. Introduction

Vibration measurement is used in various industrial fields, particularly for automobile parts whose performance must be evaluated in advance by vibration testers for safe driving. Traditionally, multiple contact sensing devices, such as accelerometers, have been used to measure vibrations [1]. To overcome the problems of contact sensing devices, such as target interference, limited measuring positions, and difficult installation, digital image correlation (DIC) is widely used for the non-contact measurement of dynamic features, such as two-dimensional (2D) displacement and 2D strain [2,3]. DIC has been extended to acquire three-dimensional (3D) dynamics using two or more cameras. However, it is not robust when faced with bad patterns because of the pixel-level correspondence between the left and right images that requires pattern-based searching and matching in pursuit of accuracy. And performing a local analysis between different measurements is almost impossible. To overcome these limitations, a model-based 3D vibration measurement method based on DIC using stereo high-speed cameras is proposed in this study. Establishing the relationship between the 3D model and images in advance by treating the 3D model as a medium, so that points from left and right images can be matched directly through the 3D model without pattern-based searching. This makes processing robust. Then mapping calculated results to the 3D model makes it locally analyzable. The effectiveness of the proposed method is demonstrated through a vibration measurement experiment using a speaker.

## 2. Model-Based 3D Vibration Measurement Method

The proposed method introduces a 3D model into processing based on which image correspondence is easily realized between stereo high-speed cameras for calculating the 3D relative vibration by triangulation. This concept is illustrated in Fig. 1. Assuming that the target is a nearly rigid body and that its vibration and deflection are small, the use of the 3D model provides a strong reference for 3D-to-2D registration. Based on this, first, the correspondence between the left and right images can be easily realized and confusing matching cases can be avoided. Searching along the epipolar lines is no longer required. Second, measurements based on the same position of the 3D model enable to perform local recordings and comparisons for local analysis between different measurements. Third, external parameter calibration in advance is no longer necessary, particularly in situations where calibrating the stereo high-speed cameras is difficult.

## 3. Vibration Measurement Experiment on a Speaker

### *3.1 Configuration*

Fig. 2 shows the experimental configuration, consisting of left and right high-speed cameras (Mikrotron, EoSens 4CXP), light-emitting diode (LED) lighting, and a target speaker. The high-speed cameras were capable of capturing 8-bit gray images at a maximum of 563 fps with a full resolution of 2336 × 1728 pixels. The complementary metal-oxide-semiconductor (CMOS) image sensor size of the camera was 16.35 mm × 12.10 mm, with a pixel size of 7 μm × 7 μm. The specifications of the desktop personal computer (PC) used for image capture and processing were as follows: Main Board: MPG Z690 CARBON WIFI (MS-7D30), CPU: 12th Gen Intel (R) Core (TM) i9-12900K, 3.19 GHz, Memory: 64 GB RAM, GPU: NVIDIA GeForce RTX 3090, and OS: Windows 11 Professional 64-bit. A 3D model of the target speaker was generated in advance using a 3D scanner (GOM, ATOS).

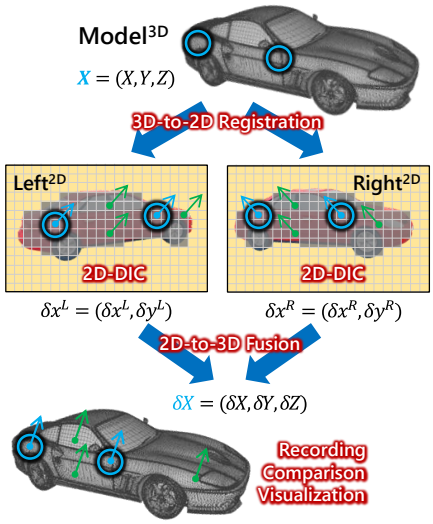



Fig.1. Concept of the proposed method

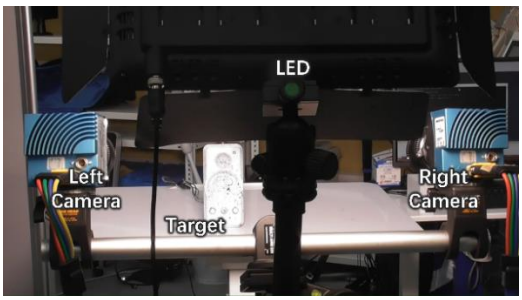
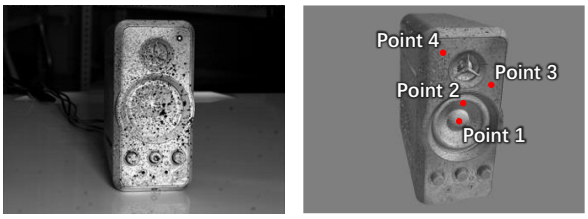



Fig.2. Experimental scene




(a) (b)

Fig.3. (a) Speaker and (b) its three-dimensional (3D) model and the selected points

*3.2 Measurement Experiment*

The left and right cameras were synchronized when capturing images around the vibration source. Fig. 3 shows the target speaker, its 3D model generated by the 3D scanner, and the points specifically selected from the 3D model for the measurements.

The measurement was implemented with a speaker playing a 100 Hz audio. A total of 2,500 frames corresponding to 500 fps for 5 s were captured by each high-speed camera and processed. The measured 3D vibrations of different points during 0 s–0.1 s are shown in Fig. 4. Totally 10 peaks are observed within the shown 0.1 s result. This corresponded to the fact that it vibrates at exactly 100 Hz. Point 1, which is located in the center of the speaker membrane, vibrates with the maximum amplitude. When it comes to the edge region, for example, point 2, the vibration amplitude decreases to 0.08 mm. Compared to the vibration of the speaker membrane, point 3 and 4 which are located in the box shell vibrate very weakly. This demonstrated that our method can accurately measure the vibration distribution across the speaker.

Fig. 5 shows the visualization in color format of 10 consecutive frames (0.054 s–0.072 s) that also refers to two cycles of vibration. The color varied from pure blue (0 mm) to pure red (0.13 mm), indicating the root sum square of the x, y, and z displacements. A comparison of the first and second rows indicated that the two frames at the same position in different cycles had similar displacements.

## 4. Conclusions

In this study, a model-based 3D vibration measurement method based on 3D-DIC using stereo high-speed cameras by introducing a 3D model was proposed. The proposed method effectively increased the robustness and ability of the local analysis. Using the proposed method, an experiment with a speaker was designed and implemented, and the recording, comparison, and visualization of the measured vibration demonstrated the effectiveness of the proposed method. In the future, we plan to implement parallel processing based on GPUs to further increase the processing speed and realize a real-time measurement method.

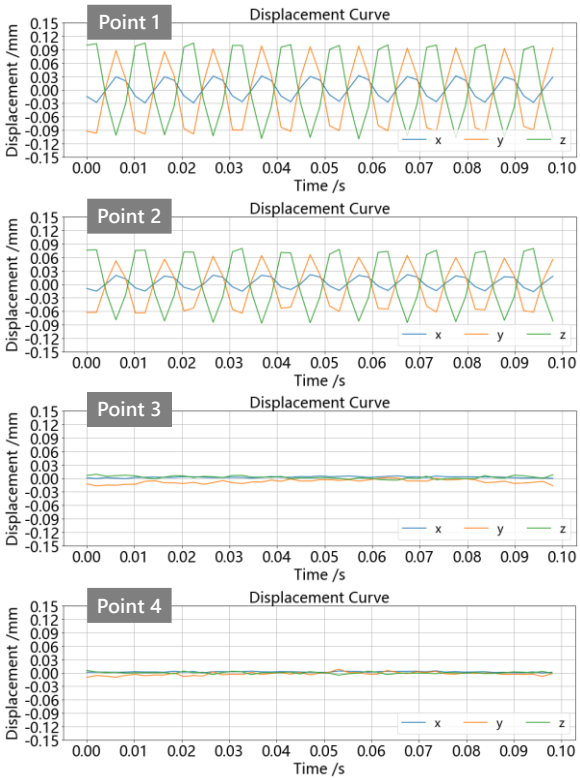



Fig.4. Measured 3D displacements of different points

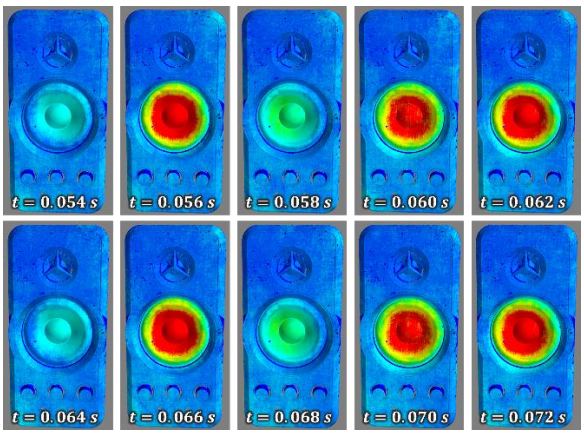



Fig.5. Visualization in color format of 10 consecutive frames that also refers to two cycles of vibration

## References

[1] A. Sabato et al.: "A novel wireless accelerometer board for measuring low-frequency and low-amplitude structural vibration," IEEE Sensors Journal, vol. 16, no. 9, pp. 2942–2949, (2016).

[2] Y. Gao et al.: "High-efficiency and high-accuracy digital image correlation for three-dimensional measurement," Optics and Lasers in Engineering, vol. 65, pp. 73–80, (2015).

[3] F. Wang et al.: "Real-time Vibration Visualization Using GPU-based High-speed Vision," Journal of Robotics and Mechatronics, pp. 1011–1023, (2022).